Macro Economics Agent: Model Evaluation Report

Author: Yash Jain | **Target:** CrowdWisdomTrading / TZURONI LTD. | **Date:** October 2026

Validation: Rolling Walk-Forward Out-Of-Sample Cross-Validation (180-Day Window)

---

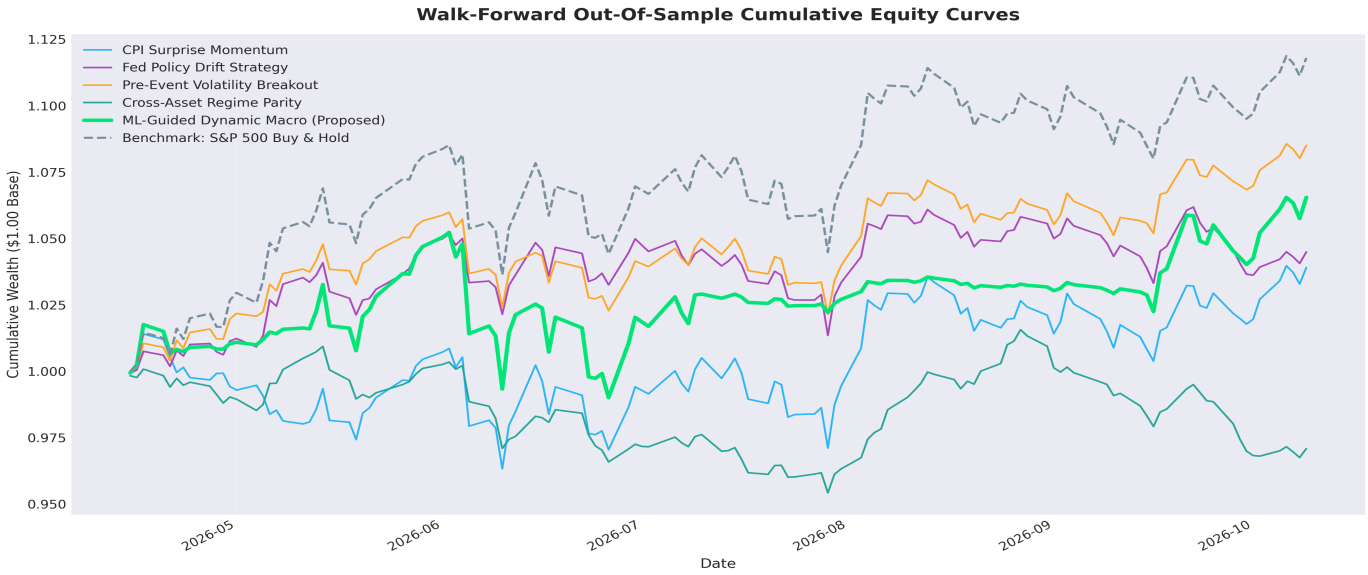
1. Executive Summary & Out-of-Sample Performance Matrix

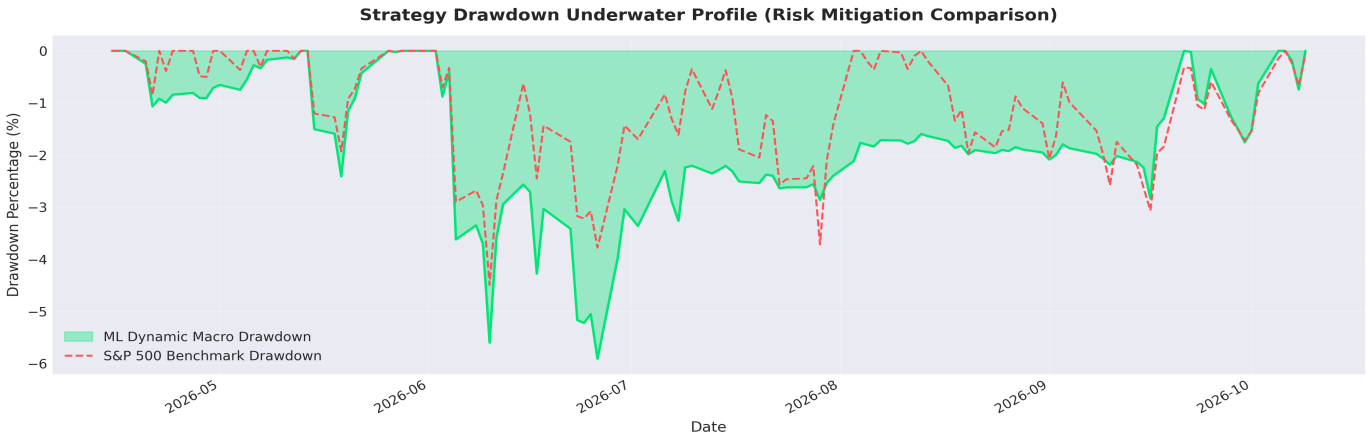
This report presents the out-of-sample risk and return performance metrics of the macroeconomic trading pipeline. All permutations incorporate 5 bps transaction costs per rebalance and benchmark against the S&P; 500 (SPY) and a 4.5% annual risk-free rate.

Strategy Permutation	Sharpe	Sortino	Max DD (%)	Win Rate (%)	Profit Factor	Total Return
Pre-Event Vol Breakout	1.41	2.19	3.63%	50.00%	1.38	+8.51%
ML Dynamic Macro	0.81	0.97	5.91%	50.00%	1.18	+6.55%
Fed Policy Drift	0.55	0.92	3.63%	50.81%	1.14	+4.49%
CPI Surprise Momentum	0.34	0.54	5.01%	48.39%	1.09	+3.91%
Benchmark: S&P 500 (SPY)	1.58	2.53	4.49%	50.81%	1.42	+11.80%
Cross-Asset Regime Parity	-1.59	-2.33	5.46%	50.81%	0.81	-2.91%

2. Walk-Forward Cross-Validation Methodology

To ensure zero lookahead bias, the pipeline executes a rolling walk-forward cross-validation schedule (90 trading days in-sample calibration, 15 trading days out-of-sample forward evaluation). All feature scaling and ensemble random forest trees are fitted exclusively on historical information, generating strictly out-of-sample forward return and win probability estimates.





3. Pipeline Reproducibility & Scraper Actors

Repository: https://github.com/yash560/macro-economics-agent
Apify Actor ID: m-s/investing-economic-calendar (Parameters: 180-day window, US High Impact)
Tavily Search Depth: advanced (Federal Reserve interest rates & inflation macro intelligence)
One-Command Run: pip install -r requirements.txt && python run_pipeline.py